Mohamed Helmy

Android & Kotlin Multiplatform Engineer

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SUMMARY

Android & Kotlin Multiplatform Engineer specializing in scalable cross-platform architectures, modular systems, and published open-source libraries. Experienced in Jetpack Compose, Compose Multiplatform, modular architecture (24+ modules), offline-first synchronization, and secure mobile systems. Proven track record of modernizing legacy codebases, accelerating build and networking pipelines, and delivering production-ready solutions.

EXPERIENCE

Areeb Technology Jul 2025 – Present
Android Software Engineer Cairo, Egypt

- Led KMP research and validated shared business logic with native UI integration under Clean Architecture, establishing a production-ready cross-platform blueprint adopted by multiple teams.
- Decomposed a monolithic codebase into 24 independent Gradle modules, cutting clean build times by 30% and virtually eliminating cross-team merge bottlenecks across parallel feature branches.
- Standardized multi-module dependency governance using Gradle Version Catalogs, adopted Navigation 3 for type-safe routing, and optimized Gradle build caching to cut incremental compile times by an additional 15%.
- Strengthened app security and compliance by replacing legacy NDK/JNI cryptography with Google Tink and redesigning encrypted DataStore architecture.
- Reduced JSON serialization overhead and client response latency by 20% by migrating the networking layer from Retrofit/Gson to Ktor with Kotlinx Serialization and connection pooling.
- Architected and shipped core production flows for Boxooo (logistics & delivery) and Cadaa (e-commerce marketplace)—including onboarding, KYC verification, biometric authentication, password recovery, push notifications, and full RTL/Arabic localization.

National Telecommunication Institute (NTI) — Bank Misr Program Oct 2026 – Present
Software Engineering & Kotlin Instructor (Part-Time) Cairo, Egypt

- Deliver an intensive curriculum covering Clean Architecture, SOLID principles, and 16+ Gang of Four (GoF) Creational, Structural, and Behavioral design patterns.
- Train software engineers on modern Kotlin, emphasizing object-oriented modeling, functional programming paradigms, structured concurrency with Coroutines and Flow, and null-safe system design.

Freelance Feb 2025 – Jul 2025
KMP Software Engineer Remote

- Engineered a real-time, offline-first POS and ERP platform for Android tablets and Desktop built on Kotlin Multiplatform, keeping multi-warehouse inventory synchronized across connectivity drops via Room and Supabase.
- Implemented secure RPC-based stock transfer and inventory workflows to maintain data consistency across multiple warehouse locations.
- Built automated CI/CD deployment pipelines using GitHub Actions for multi-platform test suites, artifact signing, and continuous delivery.

Egyptian Armed Forces Oct 2023 – Dec 2024
Military Service (Completed) Cairo, Egypt

- Completed mandatory military service.

EDUCATION

Faculty of Computer Science and Artificial Intelligence, Benha University 2026 – Present
M.Sc. in Computer Science (In Progress) Benha, Egypt

Faculty of Science, Benha University Sep 2019 – Jun 2023
B.Sc. in Computer Science, GPA: 3.24/4.0 Benha, Egypt

TECHNICAL SKILLS

Languages & Frameworks : Kotlin, Kotlin Multiplatform, Jetpack Compose, Compose Multiplatform, Swift, SwiftUI, Coroutines, Flow, StateFlow

Architecture & Patterns : Clean Architecture, GoF Design Patterns, MVVM, MVI, Repository Pattern, Dependency Injection (Hilt, Koin), Modular Architecture

Data & Networking : Room, DataStore, Ktor, Retrofit, REST, Supabase, Firebase, SQLite, Protobuf, JWT

Testing & Quality : JUnit, Mockito, Mokkery, Compose UI Testing, Screenshot Testing, Maestro, Espresso, TDD

DevOps & Tools : Git, GitHub, GitHub Actions, CI/CD, Gradle, Gradle Version Catalogs, ProGuard/R8, Benchmarking

Security & Performance : Google Tink, Encrypted DataStore, Memory Optimization, App Startup Optimization, Performance Profiling

PROJECTS

Safar | *Kotlin Multiplatform, Compose Multiplatform, Supabase*

- Built a delivery operations platform for Android and Desktop that digitizes driver dispatch, invoicing, and payroll, replacing manual spreadsheet-based workflows.
- Developed Supabase Edge Functions for automated payroll logic and implemented Row-Level Security for granular role-based access control.
- Designed an automated operations platform targeting an estimated 90% reduction in manual dispatch and billing overhead through automated invoice tracking and live driver management.

hijri-date-picker | *Kotlin Multiplatform* | Maven Central

- Developed a KMP Hijri calendar picker supporting Android, iOS, and Desktop with platform-consistent behavior.
- Shipped a reusable cross-platform library published to Maven Central (`io.github.helmy2:hijri-date-picker`) with comprehensive documentation and sample apps.
- Engineered custom rendering logic for accurate Hijri date calculations with timezone awareness and localization support.

CrossCart | *Kotlin Multiplatform, Compose Multiplatform, Supabase*

- Built an e-commerce platform that lets merchants run a complete storefront (catalog, cart, payments, and order tracking) on Android, iOS, and Desktop from a single codebase.
- Applied clean MVI architecture with unidirectional data flow and live data synchronization via Supabase.
- Designed responsive UI components adapting to different screen sizes and form factors across mobile and desktop platforms.
- Integrated payment processing, cart management, and order tracking with offline support so sales continue even without connectivity.

CERTIFICATIONS

Building Industry-Level Compose Multiplatform Apps, PL-Coding, 2026
Android Developer Track, Digital Egypt Builders Initiative (DEBI), 2025
Android Application Developer Track, Information Technology Institute (ITI), 2023
Advanced Android Nanodegree, Udacity and FWD, 2022
Machine Learning Specialization, DeepLearning.AI (Coursera), 2022
CS50: Computer Science, Harvard University, 2021
Algorithmic Toolbox, Coursera, 2021